

Torque

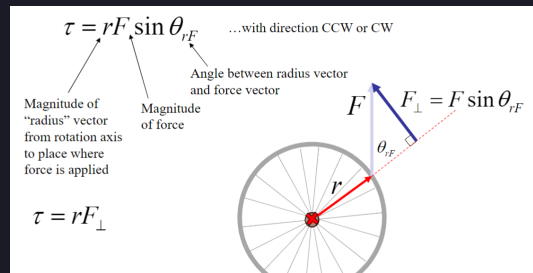
Torque is the rotational analog of force.

Angular momentum being conserved requires that net torque = 0.

$$\tau = rF \sin \theta_{rF}$$

with direction CCW or CW

Torque = the force parallel to the tangent of the circle * radius away



$$F_{\perp} = F \sin \theta$$

$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$\tau = I\alpha$$

$$\alpha = \frac{d}{dt}\omega$$

We can compute this with determinants. For example, take a radius vector going straight up, and a torque acting along the x axis at a height $h/2$ along a tree

$$\tau_1 =$$

$$\det \begin{bmatrix} \hat{x} & \hat{y} & \hat{z} \\ 0 & 0 & \frac{h}{2} \\ F_1 & 0 & 0 \end{bmatrix}$$

$$= \frac{h}{2} F_1 \hat{y}$$